

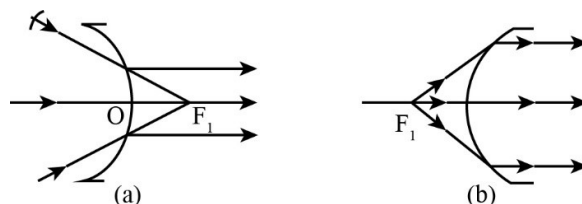
Refraction At Spherical Surfaces

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{r}$$

where u = object distance, v = image distance, r = radius of curvature

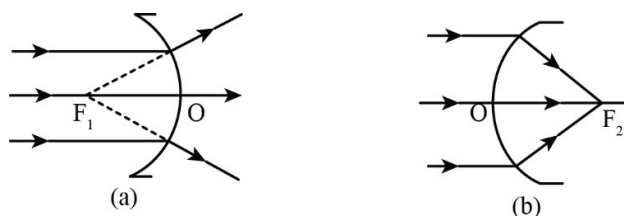
This equation is known as Gaussian formula for a single refracting surface. Here is called the $\frac{n_2 - n_1}{r}$ power of the refracting surface.

Principle Foci of a Spherical Surface



The first focal point (F_1) is a point on the principal axis such that rays moving towards that point [Figure above] or diverging from that point [Fig. above] become parallel to the principal axis after refraction. The distance of F_1 from the pole of the refracting surface is called the first principal focal length (f) of the surface. Hence $f' = u$ when $v = \infty$. From the general formula (1)

$$f = -\frac{n_1 r}{n_2 - n_1} \dots \dots (2)$$



The second focal point (F_2) is a point on the principal axis such that the incident rays moving parallel to the principal axis will, after refraction, either appear to diverge from this point [Figure above]) or actually converge to this point-[Figure above]]. This distance of F_2 from the pole of the refracting surface is called the second principal focal length (f) of the surface. Hence $f = v$ when $u = \infty$. From the general formula (1).

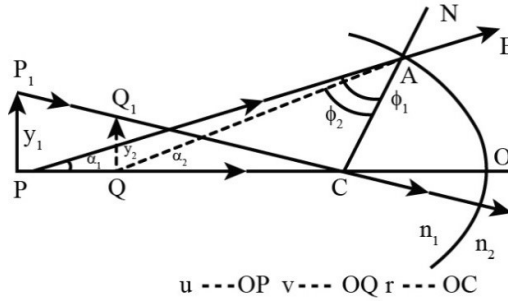
$$f = \frac{n_2 r}{n_2 - n_1} \dots \dots (3)$$

we get,

$$\frac{f}{f'} = -\frac{n_2}{n_1} \dots \dots (4)$$

This indicates that the two principal focal lengths are always of opposite signs.

Magnification for A Spherical surface:



In Figure above let PP1 be an object and QQ1 be its image formed by refraction at the concave spherical surface separating two media of refractive indices n_1 and n_2 .

Lateral or linear magnification:

Linear magnification (m) is defined as the ratio of the image height (y_2) to the object height (y_1) both measured perpendicular to the axis. Thus

$$m = \frac{y_2}{y_1} = \frac{OQ-OC}{OP-OC} \dots\dots(5)$$

Now according to our sign conventional distances measured to the left are negative. Let the object distance $u = -OP$, image distance $v = -OQ$ and the radius of curvature $r = OC$.

$$m = \frac{v-r}{u-r} \dots\dots(6)$$

$$n_1 \left(\frac{1}{r} - \frac{1}{u} \right) = n_2 \left(\frac{1}{r} - \frac{1}{v} \right) \Rightarrow n_1 \frac{u-r}{u} = n_2 \frac{v-r}{v} \dots (7)$$

Hence from Equation (6) and (7), $m = \frac{n_1}{n_2} \cdot \frac{v}{u} \dots\dots(8)$

Longitudinal magnification:

If a small object of length δu , placed along the axis produces an images of length δv , the lim

$\frac{\delta v}{\delta u} = \frac{dv}{du}$ is a measure of longitudinal magnification (l).

Differentiating the equation, $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2-n_1}{r}$ we get,

$$\begin{aligned} -\frac{n_2}{v^2} \cdot \frac{dv}{du} + \frac{n_1}{u^2} &= 0 \Rightarrow \frac{dv}{du} = \frac{n_1}{n_2} \cdot \frac{v^2}{u^2} \\ \Rightarrow l = \frac{dv}{du} &= \left(\frac{n_1 v}{n_2 u} \right)^2 \cdot \frac{n_2}{n_1} = m^2 \cdot \frac{n_2}{n_1} \dots\dots(9) \end{aligned}$$

Angular magnification:

If the incident ray PA and the corresponding refracted ray AB make angles α_1 and α_2 respectively with the axis (Figure above) the ratio $\frac{\tan \alpha_2}{\tan \alpha_1}$ is called the angular magnification (M).

$$\Rightarrow M = \frac{\tan \alpha_2}{\tan \alpha_1} = \frac{AO/OQ}{AO/OP} = \frac{OP}{OQ} = \frac{u}{v} \dots (10)$$

Using Equation (8) we can write, $M = \frac{n_1}{n_2} \cdot \frac{1}{m} \dots (11)$

Inter-relation:

Combining Equation (9) and (11) we get $l = m \times \frac{1}{M}$, $\frac{l \times M}{m} = 1$

$$\Rightarrow \frac{\text{Longitudinal magnification} \times \text{Angular magnification}}{\text{Linear magnification}} = 1 \dots (12)$$

Lagrange's Law and Helmholtz's Relation:

In the paraxial region there exists an important relation between object size, image size and the angles which the conjugate rays make with the principal axis. Let in Figure above PP_1 be a small object of size y_1 placed transversely to the axis in front of a certain refracting surface C is the centre of curvature and QQ_1 is the image of size y_2 . The conjugate rays PA and AB make angles α_1 and α_2 respectively with the axis. From Figure above we can write,

$$\frac{y_2}{y_1} \cdot \frac{\tan \alpha_2}{\tan \alpha_1} = \frac{QQ_1}{PP_1} \times \frac{AO/OQ}{AO/OP} \quad [\text{in the paraxial region the point A is very close to pole O}]$$

According to our sign convention all distances measured to the left of O are negative. Let the object distance $u = -OP$, image distance $v = -OQ$ and radius of curvature $r = -OC$.

Therefore,

$$\frac{y_2}{y_1} \cdot \frac{\tan \alpha_2}{\tan \alpha_1} = \frac{v-r}{u-r} \times \frac{u}{v} \dots (13)$$

Now we have the relation

$$\begin{aligned} \frac{n_2}{v} - \frac{n_1}{u} &= \frac{n_2 - n_1}{r} \Rightarrow n_1 \left(\frac{1}{r} - \frac{1}{u} \right) = n_2 \left(\frac{1}{r} - \frac{1}{v} \right) \\ \Rightarrow \frac{n_1}{n_2} &= \frac{v-r}{u-r} \cdot \frac{u}{v} \dots (14) \end{aligned}$$

Combining Equation (13) and (14), we get

$$n_1 y_1 \tan \alpha_1 = n_2 y_2 \tan \alpha_2$$

In general, $n_1 y_1 \tan \alpha_1 = n_m y_m \tan \alpha_m \dots (15)$

Its importance lies in the fact that it expresses linear magnification in terms of angular magnification. Note that this relationship between the magnifications is independent of the refractive indices of the intervening media.

A system is known as telescopic system if all the cardinal points lie at infinity. In this case, usually $n_i = n_{\dots}$. Hence we get

$$\frac{y_1}{y_m} = \frac{\tan \alpha_m}{\tan \alpha_1} \dots (16)$$

Thus, in a telescopic system angular magnification ($\tan \alpha_m / \tan \alpha_1$) is equal to the reciprocal of linear magnification (y_m / y_1)

ABBE'S Sine Condition:

$$n_1 y_1 \sin \alpha_1 = n_2 y_2 \sin \alpha_2$$

This is known as Abbe's sine condition. It holds good for all rays whether paraxial or not.

When the rays are paraxial i.e α_1 and α_2 are small we can write $\sin \alpha_1 \approx \tan \alpha_1$ and $\sin \alpha_2 \approx \tan \alpha_2$. In this case Abbe's sine condition reduces to Helmholtz's relation.

Refraction Through Thin Lens:

Some important terms:

(i) Principal axis: Principal axis of a lens is the straight line joining the centres of curvature of its two bounding surfaces.

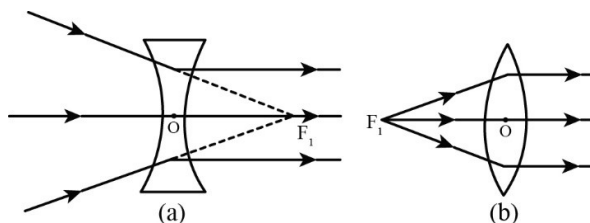
(ii) Poles: The points of intersection of the two surfaces with the principal axis are called the poles of the lens. The distance between the two poles is called the thickness of the lens (which might be considered zero for thin lens),

(iii) Principal Section: The section of lens by a plane through its principal axis is called principal section.

(iv) Aperture: When the bounding line of the lens is circular, the diameter of this circular boundary is called aperture of the lens.

(v) Optical centre: Optical centres of a lens is a point on the principal axis such that any ray through it emerges parallel to its direction of incidence. For equi-convex lens the optical centre is situated at the mid-point.

(vi) First focal point: first focal point of a lens is a point on the principal axis such that rays moving towards that point [Figure below] or diverging from that point [Figure below] become parallel to the principal axis after refraction. The distance of F_1 from the optical centre (O) of the lens is called the first focal length (f'), when $v = \infty$.



(vii) Second focal length: Second focal length (F_2) is a point on the principal axis such that the incident rays moving parallel to the principal axis will, after refraction, either appear to diverge from this point [Figure above] or actually converge to point [Figure above]. The distance of F_2 from the optical centre (O) of the lens is called the second focal length (f). Hence $f = v$ when $u = \infty$. According to our sign convention f is negative for a diverging lens and positive for a converging lens.